

DEPARTMENT OF MINES

SOUTH AUSTRALIA

MAGNETIC OBSERVATIONS IN THE MIDDLEBACK RANGES, 1951

GEOPHYSICAL REPORT 4/51

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by

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MICROFILMED

Further to the observations carried out by the writer in this area, observations have been made along a number of traverses suggested by the party carrying geological mapping in this area, under the supervision of Dr. K.R. Miles.

These reconnaissance traverses were at widely spaced intervals the relative positions of which were fixed, as in the previous survey by the use of aerial photographs.

Results.

The readings taken on these traverses have been plotted, as before, on the preliminary geological maps of the area.

Along the traverses to the west of the main Camel Hill line, where it has been proposed that diamond drilling should be carried out to determine possible extensions of the banded iron formation, there are no magnetic anomalies of comparable magnitude to those over the known formation.

Therefore, if the steeply dipping beds of the Camel Hill line do return to the surface in this area, it is at a very low angle of dip.

It is thought probable that the trough of low readings extending in a north west direction from the northern end of the gap in the range may be associated with the end of these beds, which would thus appear to have been cut off at considerable depth, except on the most northern of this set of traverses, where an anomaly of 1000 gammas would appear to be associated with a remnant of the formation.

In the area to the north of Iron Knob township the traverses to date indicate that the iron formation extends about one mile north of the township where there is still a marked anomaly, but that beyond this point the large anomaly disappears, probably due to the formation pitching downwards towards the north.

The only other traverses on which anomalies of the order found over the magnetite bearing iron formation were observed were on the one between the north end of the Iron Duchess and the south end of the Iron Knight, and the most southerly traverse through the gap in the Camel Range line to the west of Mt. Middleback North.

N.B. It should be remembered that the large magnetic anomalies are due to the magnetic content of the beds, and that should the magnetite be sufficiently oxidised there may be no such anomaly e.g. in the taconite beds in the Lake Superior district the less magnetic parts generally correspond to the enriched ore zones.

Further work to the north of Iron Knob may be useful to follow the trend of the iron formation, but it is considered that as an aerial magnetic reconnaissance is to be carried out in the near future, this survey become available as the work can then be better directed.

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